QP CODE: 25900241

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Value Addition Course (VAC)-MG3VACBIF200 - ENVIRONMENTAL SCIENCE AND HUMAN RIGHTS (2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest (1) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions. Answer any 5 questions
Each question carries 2 marks

1. Discuss how Liebig and Shelford law help in understanding organisms distribution. [U]/[CO1]
   - Liebig's Law of Minimum states that an organism's distribution depends on the scarcest environmental factor. (1mark)
   - Shelford's Law of Tolerance adds that organisms can survive only within certain range of environmental conditions. (1 mark)
2. State difference between global warming and ozone depletion. [K]/[CO2]
   - Global warming is the rise in Earth's average temperature due to increased greenhouse gases (1 mark)
   - Ozone depletion is the thinning of the ozone layer caused by chemicals like CFCs, allowing more UV radiation to reach Earth (1 mark).
3. How does the Green Data Book differ in purpose from the Red Data Book? [U]/[CO4]
   - The Red Data Book lists endangered and threatened species to help in their protection. (1mark)
   - The Green Data Book provides environmental and natural resource data to support sustainable development and conservation planning. (1 mark)
4. How would you involve local communities in rainwater harvesting practices? [A]/[CO4]
   - Education programs, community participation, awareness campaigns, training, capacity building, forming local groups to manage water, demonstration (any three points 2 marks)
5. How do the Directive Principles of State Policy (DPSP) support Human Rights in India? [U]/[COS]
   - It ensures social, economic, and political justice. They promote equality, education, health, and livelihood, helping to secure a dignified life for all citizens. (1 mark for each relevant point)
6. Evaluate effects of deforestation. [An]/[CO3]
   - Loss of biodiversity, soil erosion, disruption of the water cycle, floods, lesser rains, loss of livelihood etc (2 marks for any 3 relevant points)

7. Recall difference between desert and forest ecosystem. [K]/[C01]
   - A desert ecosystem has very low rainfall, sparse vegetation, and organisms adapted to dry conditions. (1 mark)
   - A forest ecosystem has high rainfall, dense vegetation, and rich biodiversity with many plant and animal species. (1 mark)

[2x5 = 10]

Part B

Short Essay Questions. Answer any 4 questions
Each question carries 5 marks

8. Recall species richness in terms of alpha diversity. [K]/[CO2]
   - Definition of alpha diversity and explanation of species richness – 2 marks
   - Relation between alpha diversity and species richness with example and its importance – 3 marks

9. Infer any nuclear accidents. [An]/[CO3]
   - Naming the nuclear accident along with date and location – 2 marks
   - Cause of the accident(s) along with its effects – 2 marks
   - Significance / lesson learned (e.g., need for safety measures) – 1 mark

10. Write a note on species diversity. [A]/[CO4]
    - Definition of species diversity, species richness – 2 marks
    - Importance/role in ecosystem stability – 2 marks
    - Example of a diverse ecosystem – 1 mark

11. Write a short note on the historical development of Human Rights. [K]/[CO5]
    - Introduction of human rights – 1 mark
    - Mention of Magna Carta – 1 mark
    - Mention of French Revolution – 1 mark
    - Mention of UN & UDHR – 1 mark
    - Importance/impact of these developments – 1 mark

12. Identify why landscape alterations considered as an environmental issue. [U]/[CO3]
    - Definition of landscape alteration with example– 2 marks
    - Impact on biodiversity and overall environmental significance – 3 marks

13. Recall the role of solar energy in the process of energy flow. [K]/[CO1] [5x4 = 20]
    - Solar energy as primary energy source with overall importance– 2 marks
    - Role in photosynthesis Production of food– 2 marks
    - Transfer of energy to herbivores and higher trophic levels – 1 mark

[5x4 = 20]

Part C

Long Essay Questions. Answer any 2 questions.
Each question carries 10 marks

14. Explain the importance of population density in ecosystem balance. [U]/[CO2]

- Definition of population density – 1 mark
- Importance for resource availability – 2 marks
- Role in predator-prey balance – 2 marks
- Impact on disease spread and genetic diversity – 2 marks
- Effect on competition – 2 marks
- Contribution to ecosystem productivity and balance – 1 mark

15. Differentiate various types of pollution. [An]/[CO3]

- Air pollution – source + effect – 2 marks
- Water pollution – source + effect – 2 marks
- Soil pollution – source + effect – 2 marks
- Noise pollution – source + effect – 1 mark
- Light pollution – source + effect – 1 mark
- Thermal pollution – source + effect – 1 mark
- Radioactive pollution – source + effect – 1 mark

16. What is a National Park? Describe its characteristics and give examples from India. [U]/[CO4]

- Definition of National Park with examples – 3 marks
- Legal protection / purpose – 2 marks
- Characteristics - biodiversity, habitat preservation, limited human interference, tourism regulation – 5 marks

17. Write an essay on the objectives and key provisions of the Biodiversity Act, 2002. [K]/[CO5]

- Introduction / Purpose of the Act – 2 marks
- Objectives: conservation, sustainable use, benefit sharing, regulation of access – 4 marks
- Key Provision 1 (NBA, SBB, BMCs) – 1 mark
- Key Provision 2 (access regulation) – 1 mark
- Key Provision 3 (benefit sharing & documentation) – 1 mark
- Key Provision 4 (penalties) – 1 mark

[10x2 = 20]